Make/Model Assignment in UrbanEV–Maryland

A code-level walkthrough: data sources, sampling, attributes, and downstream uses

UrbanEV–Maryland documentation series

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1 Overview

Every synthetic EV owner receives a specific vehicle archetype ("make/model"), which determines battery capacity, energy consumption, fast-charging network access, the counterfactual gasoline vehicle used for the shadow-tax computation, and, for plug-in hybrids, the per-kWh cost of the gasoline fallback. Assignment happens in `pipeline/scripts/s6_plans/06_build_plans.py` (the build that produced all reported results), drawing on one reference table and one MATSim vehicle-types file. The chain is: MVA registrations → powertrain draw (script 04) → archetype draw within powertrain (script 06) → per-agent attributes → MATSim fleet file and person attributes.

2 Data sources, provenance, and citations

Tracing the code revealed a two-tier provenance that must be stated precisely, because the two tiers have different authority.

2.1 County × powertrain counts: official MVA registrations

The file `pipeline/data/reference/mva/MDOT_MVA_Electric_and_Plug-in_Hybrid_Vehicle_Registrations_b` is the official MDOT MVA dataset (columns: `Year_Month, Fuel_Category, County, Count`). It governs the county ownership calibration targets, the statewide total (148,359), and each county's BEV/PHEV split (statewide 73.8% BEV).

Citation: Maryland Department of Transportation, Motor Vehicle Administration. "MDOT MVA Electric and Plug-in Hybrid Vehicle Registrations by County (as of each month end)." Maryland Open Data Portal, `https://opendata.maryland.gov`. Snapshot retrieved March 13, 2026; January-2026 month used.

2.2 Make/model shares within powertrain: national sales-mix estimates

The `fleet_count/fleet_share_pct` columns of the archetype lookup sum to 99,132 — the composition of the upstream synthetic fleet, whose make/model shares were author-estimated from publicly reported 2025 U.S. EV sales tracking, with within-tier renormalization (the upstream script documents this explicitly: "The shares below are author-estimated from publicly reported 2025 US EV sales tracking (Cox Automotive / Kelley Blue Book quarterly reports, InsideEVs monthly tracker). They are NOT authoritative MD-registration figures."). Maryland's MVA county dataset does not include make/model detail, so the national mix is a proxy.

Citations: Cox Automotive / Kelley Blue Book. Quarterly U.S. electric-vehicle sales reports,

2025. https://www.coxautoinc.com. InsideEVs. Monthly U.S. EV sales scorecard, 2025. https://insideevs.com/news/category/sales.

2.3 Vehicle specifications: EPA fueleconomy.gov

Battery capacity, consumption, the counterfactual ICE match, its EPA combined mpg, and the PHEV charge-sustaining mpg are sourced row-by-row from EPA vehicle pages; every row of the lookup carries its `source_url` and `retrieval_date` (2026-06-07), and 49 of 57 rows covering 93.3% of the fleet are marked `verified` (long-tail buckets use segment-typical values).

Citation: U.S. Department of Energy and U.S. Environmental Protection Agency. Fuel Economy Guide vehicle pages, model years 2024–2025. https://www.fueleconomy.gov. Retrieved June 7, 2026.

2.4 Gasoline prices for the PHEV fallback: AAA

Per-archetype gasoline fallback costs divide the AAA Maryland average pump price by the EPA charge-sustaining mpg (premium-fuel models use the premium price).

Citation: American Automobile Association. "Maryland Average Gas Prices." https://gasprices.aaa.com/?state=MD. Retrieved July 16, 2026 (regular $3.881/gal, premium $4.805/gal).

Provenance correction to the manuscript

Because of the two-tier structure, the phrase "make/model drawn from Maryland registration shares" previously used in the manuscript overstates the second tier: only the county powertrain mix is MVA-sourced; within-powertrain model shares are the national 2025 sales mix. The manuscript and figure captions should read "drawn from the 2025 U.S. EV sales mix, with county powertrain shares from MVA registrations."

3 Step 1: Powertrain draw (script 04)

In `04_ev_ownership_cirillo.py`, after county-calibrated ownership sampling, each owner's powertrain is a Bernoulli draw on the county's MVA BEV share:

```
o["ev_powertrain"] = np.where(rng.random(len(o)) < mva_bev.get(fips, 0.738),
                              "BEV", "PHEV")
```

so counties keep their observed BEV/PHEV mix and the statewide share converges to the MVA value of 73.8% BEV.

4 Step 2: Archetype draw within powertrain (script 06)

The lookup is split by powertrain and each owner samples one archetype with probability proportional to its MVA sales-mix share, normalized within the powertrain:

```
for pt in ["BEV", "PHEV"]:
    sub = look[look.powertrain == pt]
    veh[pt] = (sub.ev_type.to_numpy(),
               (sub.fleet_share_pct / sub.fleet_share_pct.sum()).to_numpy(),
               dict(zip(sub.ev_type, sub.battery_kwh)))
for pt in ["BEV", "PHEV"]:
    m = (ev.ev_powertrain == pt).to_numpy()
```

```
    types, probs, batt = veh[pt]
    pick = rng.choice(len(types), m.sum(), p=probs)
    ev_type[m] = types[pick]; battery[m] = [batt[types[i]] for i in pick]
```

The generator is seeded (`np.random.default_rng(SEED)`), so the fleet is reproducible. In expectation the assigned composition equals the source sales-mix shares exactly; realized shares differ only by multinomial noise.

5 Step 3: Attributes that follow from the archetype

- Battery capacity: `battery_kwh` from the lookup, written to the fleet file per vehicle.
- Initial state of charge: BEVs draw from a Beta(4,2) clipped to [0.20, 0.95] (most cars start fairly full); PHEVs draw uniform [0.30, 0.90].
- Fast-charging access: the Tesla set {`model_s`, `model_3`, `model_x`, `model_y`, `cybertruck`} receives `charger_types="L1,L2,DCFC,DCFC_TESLA"`; all other archetypes receive `"L1,L2,DCFC"`. This is why 100% of simulated Tesla Supercharger sessions belong to Tesla models.
- PHEV correction (patch): 18 of the 21 PHEV archetypes originally carried L2-only capability; `patch_plans_phev.py` normalized them to `"L1,L2"` (PHEVs do not fast-charge), and the same patch attached the per-archetype `phevGasCostPerKwh` attribute.
- PHEV gasoline fallback price: computed per archetype by `make_phev_gas_costs.py` as AAA Maryland pump price divided by the EPA charge-sustaining mpg, converted to an equivalent \$/kWh (fleet mean \$0.364/kWh, range \$0.21–0.56); premium-fuel models use the premium price.
- Counterfactual mpg: the archetype's segment-matched ICE mpg feeds the shadow gas-tax gap R^* and the gas-tax-benchmark incidence, so a Cybertruck owner's counterfactual pays the fuel tax of an F-150, not of an average car.

The archetype key is written both as the person attribute `evModel` (used by all analysis scripts) and as the vehicle's `vehicle_type` (used by the simulator's consumption model):

```
pf.write(A("evModel", "String", row.ev_type))
vf.write(f'<vehicle id=... battery_capacity="{battery[g]:.2f}"
  initial_soc="{soc_frac[g]*battery[g]:.2f}"
  charger_types="{charger_types[g]}" vehicle_type="{esc(row.ev_type)}" />')
```

6 Calibration status and validation

Make/model composition is an input, not a validation claim: shares are sampled from the documented sales mix, so agreement with that mix is by construction (checked in the Tier-3 fleet-composition figure; county totals match at MAPE 3.1%, $r > 0.999$). What is not imposed and therefore validates downstream: which households own which powertrain mix by county, the emergent PHEV utility factor (0.50–0.59 against the EPA rated 0.58), and every charging behavior built on these vehicles.

7 Honest limitations

1. Within a powertrain, archetype choice is independent of household income: income shapes ownership, home-charging access, and the money sensitivity, but a low-income owner is as likely to draw a Model S as a high-income owner. Vehicle-price–income matching would sharpen incidence results slightly.

2. Counterfactual vehicles are segment matches; models without an ICE sibling use a proxy (Cybertruck → F-150).

3. Within-powertrain model shares are a national 2025 sales-mix proxy, not Maryland registrations; if MVA publishes make/model detail, the lookup's share column can be swapped in place without touching any code.

4. The fleet is a static snapshot; it does not age or turn over within the simulation.